

SAFETY DATA SHEET

1. Identification

Product identifier: VISIOMER® EHMA

Chemical name: 2-ethylhexyl methacrylate

Other means of identification

Recommended use: Monomer
Chemical Intermediate
For use in industrial installations or professional treatment only.

Recommended restrictions: Applications where liquid monomer is intended to come into contact with skin or nails.

Manufacturer/Importer/Distributor Information

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2. Hazard(s) identification

Classification according to GHS

Health Hazards

Skin sensitizer Sub-category 1B

Environmental Hazards

Acute hazards to the aquatic environment Category 2

Chronic hazards to the aquatic environment Category 3

Label Elements

Hazard Symbol:



Signal Word: Warning

Hazard Statement: May cause an allergic skin reaction.
 Toxic to aquatic life.
 Harmful to aquatic life with long lasting effects.

Precautionary Statements

Prevention: Avoid breathing dust/fume/gas/mist/vapors/spray. Contaminated work clothing should not be allowed out of the workplace. Avoid release to the environment. Wear protective gloves.

Response: IF ON SKIN: Wash with plenty of soap and water. If skin irritation or rash occurs: Get medical advice/attention. Take off contaminated clothing and wash it before reuse.

Disposal: Dispose of contents/container to an appropriate treatment and disposal facility in accordance with applicable laws and regulations, and product characteristics at time of disposal.

Other hazards: Polymerization with heat evolution may occur in the presence of radical forming substances(e.g. peroxides), reducing substances, and/or heavy metal ions.

3. Composition/information on ingredients

Chemical name:
 2-ethylhexyl methacrylate

Substances

Chemical Identity	Common name and synonyms	CAS number	Content in percent (%) [*]
2-ethylhexyl methacrylate		688-84-6	>=98,0%

^{*} All concentrations are percent by weight unless ingredient is a gas. Gas concentrations are in percent by volume.

Composition information of impurities and stabilizers

Chemical Identity	Common name and synonyms	CAS number	Content in percent (%) [*]
Methyl methacrylate	No data available.	80-62-6	0,1 - <1%

^{*} All concentrations are percent by weight unless ingredient is a gas. Gas concentrations are in percent by volume.

The exact concentration has been withheld as a trade secret.

4. First-aid measures

Description of necessary first-aid measures

General information:

First aider needs to protect himself. Immediately remove contaminated clothing. Medical treatment is necessary if symptoms occur which are obviously caused by skin or eye contact with the product or by inhalation of its vapours.

Inhalation:

Move subject to fresh air and keep him calm. Seek medical advice.

Skin Contact:

In case of contact with skin wash off immediately with soap and water. If skin irritation occurs consult a physician.

Eye contact:

In case of contact with eyes rinse thoroughly with plenty of water and seek medical advice.

Ingestion:

Do not induce vomiting. Get medical attention immediately.

Personal Protection for First-aid Responders:

No data available.

Most important symptoms and effects, both acute and delayed

Symptoms:

Skin sensitizer

Hazards:

No data available.

Indication of immediate medical attention and special treatment needed

Treatment:

Treat symptomatically.

5. Fire-fighting measures

General Fire Hazards:

Keep unauthorized personnel away. Containers exposed to heat (fire) may build up pressure. Cool by splashing with water. Prevent fire extinguishing water from contaminating surface water or the ground water system. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Suitable (and unsuitable) extinguishing media

Suitable extinguishing media: foam Dry chemical. Carbon dioxide

Unsuitable extinguishing media: High volume water jet.

Special hazards arising from the substance or mixture:

May be released in case of fire: carbon monoxide, carbon dioxide, organic products of decomposition.

Special protective equipment and precautions for firefighters
Special fire fighting procedures:

Keep away from sources of ignition - No smoking. Take action to prevent static discharges. Use explosion-proof equipment. In the event of fire, cool the endangered containers with water. Vapours are heavier than air and can form an explosive mixture with air. When heated above the flash point and/or during spraying (atomizing), ignitable mixtures may form in air.

Special protective equipment for fire-fighters:

Use breathing apparatus.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures:

Assure sufficient ventilation. Use personal protective clothing. Keep away sources of ignition. Use breathing apparatus if exposed to vapours/dust/aerosol. Do not breathe vapours or spray mist. Do not eat, drink or smoke when using this product. Wash hands thoroughly after handling.

Accidental release measures:

No data available.

Methods and material for containment and cleaning up:

Larger quantities: Remove mechanically (by pumping). Use explosion-proof equipment! Smaller quantities and/or residues: Contain with absorbent material (e.g. sand, diatomaceous earth, acid absorbent, universal absorbent or sawdust). To be disposed of in compliance with existing regulations.

Environmental Precautions:

Do not discharge into the drains/surface waters/groundwater. Contain spilled product and prevent any contamination of soil, the sewer system or water bodies. If the product contaminates rivers and lakes or drains inform respective authorities.

7. Handling and storage

Handling

Technical measures (e.g. Local and general ventilation):

For monitoring procedures refer for instance to "Empfohlene Analysenverfahren für Arbeitsplatzmessungen", Schriftenreihe der Bundesanstalt für Arbeitsschutz and "NIOSH Manual of Analytical Methods", National Institute for Occupational Safety and Health

Safe handling advice:

Do not breathe vapors. Avoid contact with skin and eyes. Do not eat, drink or smoke during use. Keep container tightly closed. Provide good room ventilation even at ground level (vapours are heavier than air). Keep away from sources of ignition - No smoking. Take action to prevent static discharges. Use explosion-proof equipment. In the event of fire, cool the endangered containers with water. Vapours are heavier than air and can form an explosive mixture with air. When heated above the flash point and/or during spraying (atomizing), ignitable mixtures may form in air.

Contact avoidance measures:

No data available.

Storage

Safe storage conditions:

Fill the container by approximately 90 % only as oxygen (air) is required for stabilisation. With large storage containers make sure the oxygen (air) supply is sufficient to ensure stability. Keep only in the original container at a temperature not exceeding 30 °C. Can polymerize with intense heat release. Protect from the action of light.

Safe packaging materials:

No data available.

8. Exposure controls/personal protection

Control Parameters

Occupational Exposure Limits

Observe national threshold limit values.

Biological Limit Values

No biological exposure limits noted for the ingredient(s).

Appropriate Engineering Controls

For monitoring procedures refer for instance to "Empfohlene Analysenverfahren für Arbeitsplatzmessungen", Schriftenreihe der Bundesanstalt für Arbeitsschutz and "NIOSH Manual of Analytical Methods", National Institute for Occupational Safety and Health

Individual protection measures, such as personal protective equipment

General information: No data available.

Eye/face protection: Tightly fitting safety goggles

Skin Protection

Hand Protection:

Material: butyl rubber gloves
 Glove thickness: 0,3 mm
 Guideline: EN 374
 Additional Information: Suitable as spray protection.
 Material: nitrile rubber gloves
 Break-through time: 480 min
 Glove thickness: 0,4 mm
 Guideline: EN 374
 Additional Information: Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion, and the contact time., The above mentioned hand protection is based on special knowledge of the chemical and the intended handling of this product, however, it still may not be suited for all workplaces. A qualified hazard assessment should be made prior to the onset of work in order to determine the suitability of gloves for specific working environments and processes., Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Other: On handling of larger quantities: face mask, chemical-resistant boots and apron

Respiratory Protection: respiratory equipment in the case of high concentrations

Hygiene measures: Store work clothing separately. Immediately remove contaminated clothing. Follow the usual good standards of occupational hygiene. Clean skin thoroughly after work; apply skin cream.

9. Physical and chemical properties

Information on basic physical and chemical properties

Appearance

Physical state: liquid
Form: liquid
Color: colorless
Odor: ester-like
Odor Threshold: No data available.
Freezing point: < -58 °F/< -50 °C
Boiling Point: 441 °F/227 °C (1.013 hPa)
Flammability: Not applicable

Upper/lower limit on flammability or explosive limits

Explosive limit - upper:	No data available.
Explosive limit - lower:	No data available.
Flash Point:	Approximate 207 °F/97 °C (DIN 51755) Literature
Autoignition Temperature:	Not applicable
Decomposition Temperature:	No decomposition if used as directed.
pH:	No data available.
Viscosity	
Dynamic viscosity:	No data available.
Kinematic viscosity:	2,12 mm ² /s (68 °F/20 °C, DIN 51562)
Flow Time:	No data available.
Solubility(ies)	
Solubility in Water:	3,10 mg/l (68 °F/20 °C, OECD 105) Own study
Solubility (other):	miscible with most organic solvents
Partition coefficient (n-octanol/water):	4,95 (OECD 107)
Vapor pressure:	0,065 hPa (68 °F/20 °C)
Relative density:	No data available.
Density:	0,884 g/cm ³ (68 °F/20 °C) (DIN 51757)
Bulk density:	No data available.
Relative vapor density:	6,8 68 °F/20 °C
Other information	
Minimum ignition temperature:	482 °F/250 °C (1.013 hPa)
Impact sensitivity:	No data available.
Molecular weight:	198,3 g/mol

10. Stability and reactivity

Reactivity:	No data available.
Chemical Stability:	No decomposition if used as directed.
Possibility of hazardous reactions:	Polymerization with heat evolution may occur in the presence of radical forming substances(e.g. peroxides), reducing substances, and/or heavy metal ions.
Conditions to avoid:	The product is normally supplied in a stabilized form. If the permissible storage period and/or storage temperature is exceeded, the product may polymerize with heat evolution.
Incompatible Materials:	Reactions with strong oxidizing agents.
Hazardous Decomposition Products:	None when used as directed.

11. Toxicological information

General information: The substance is rapidly metabolized

Information on toxicological effects

Information on likely routes of exposure

Inhalation: Relevant route of exposure. Information on effects are given below.

Skin Contact: Relevant route of exposure. Information on effects are given below.

Eye contact: Relevant route of exposure. Information on effects are given below.

Ingestion: If handled correctly, not a relevant route of exposure. Information on effects are given below.

Symptoms related to the physical, chemical and toxicological characteristics

Inhalation: No specific symptoms noted.

Skin Contact: Prolonged or repeated contact may cause skin sensitization in susceptible individuals.

Eye contact: Eye may become red, tear, and become painful.

Ingestion: If handled correctly, not a relevant route of exposure. Information on effects are given below.

Acute toxicity (list all possible routes of exposure)

Oral

Product: LD 50 (Rat): > 2.000 mg/kg Not toxic after single exposure; Limit Test, No deaths observed.

Components:

2-ethylhexyl methacrylate No data available.

Methyl methacrylate LD 50 (Rat): > 5.000 mg/kg

Dermal

Product: LD 50 (Guinea Pig): > 17.620 mg/kg

Components:

2-ethylhexyl methacrylate LD 50 (Guinea Pig): > 17.620 mg/kg

Methyl methacrylate LD 50 (Rabbit): > 5.000 mg/kg

Inhalation

Product: LC 50: 5000 - 11000 ppm

Components:

2-ethylhexyl methacrylate No classification, Vapour Not applicable, Dust and mist

Methyl methacrylate LC 50 (Rat, 4 h): 29,8 mg/l Vapour Dust and mist, No data available.

Repeated dose toxicity

Product: NOAEL : 120 mg/kg
Components:
 2-ethylhexyl methacrylate NOAEL : 120 mg/kg
 Methyl methacrylate NOAEL (Rat, Inhalation(Vapour)): 25 ppm
 NOAEL (Rat, Oral): 2000 ppm

Skin Corrosion/Irritation

Product: OECD 439 Not irritating; in vitro assay
 Irritating.; EU-CLP as per Regulation (EU) No. 1272/2008, Annex VI
Components:
 2-ethylhexyl methacrylate OECD 439 (Human, reconstructed epidermis (RhE) model): Not irritating
 Methyl methacrylate (Rabbit): Irritating.

Serious Eye Damage/Eye Irritation

Product: Draize (Rabbit): Not irritating;
 Irritating.; EU-CLP as per Regulation (EU) No. 1272/2008, Annex VI
Components:
 2-ethylhexyl methacrylate Draize (Rabbit): Not irritating
 Methyl methacrylate Draize (Rabbit): Not irritating

Respiratory or Skin Sensitization

Product: (Guinea Pig)Skin sensitizer
 In animal experiments the substance shows low resp. no ability as a sensitizer. Skin Sensitisation Category 1B (UN-GHS)
Components:
 2-ethylhexyl methacrylate Maximization Test (Guinea Pig): Skin sensitizer
 Methyl methacrylate Local Lymph Node Assay (LLNA), OECD 429 (Mouse): Skin sensitizer

Carcinogenicity

Product: An Expert Judgment stated that no classification is necessary based on present knowledge.
Components:
 2-ethylhexyl methacrylate No data available.
 Methyl methacrylate Not classified

Germ Cell Mutagenicity

Not classified

In vitro

Product: (Ames test)no evidence of mutagenic effects; In-vitro experiments have shown no mutagenic/genotoxic effect.;

Components:
 2-ethylhexyl methacrylate (Ames test)no evidence of mutagenic effects In-vitro experiments have shown no mutagenic/genotoxic effect.
 Methyl methacrylate No data available.

In vivo
Product: No data available.

Components:
 2-ethylhexyl methacrylate No data available.
 Methyl methacrylate No data available.

Reproductive toxicity
Product: An Expert Judgment stated that no classification is necessary based on present knowledge.

Components:
 2-ethylhexyl methacrylate No data available.
 Methyl methacrylate Not classified

Specific Target Organ Toxicity - Single Exposure
Product: Not classified no evidence for hazardous properties

Components:
 2-ethylhexyl methacrylate Not classified
 Methyl methacrylate Inhalation - vapor: Respiratory system - Category 3 with respiratory tract irritation.

Specific Target Organ Toxicity - Repeated Exposure
Product: Not classified no evidence for hazardous properties

Components:
 2-ethylhexyl methacrylate no evidence for hazardous properties
 Methyl methacrylate Not classified

Aspiration Hazard
Product: Not classified no evidence for hazardous properties (structure-activity-relationships) (analogy)

Components:
 2-ethylhexyl methacrylate Not classified
 Methyl methacrylate Not classified

Information on health hazards
Other hazards
Product: Avoid contact with the skin and eyes and inhalation of the product vapours.;

12. Ecological information

Ecotoxicity:
Acute hazards to the aquatic environment:
Fish
Product: LC 50 (Oryzias latipes (Orange-red killifish), 96 h): 2,78 mg/l

Components:
 2-ethylhexyl methacrylate LC 50 (Oryzias latipes (Orange-red killifish), 96 h): 2,78 mg/l
 Methyl methacrylate LC 50 (Oncorhynchus mykiss, 96 h): > 79 mg/l

Aquatic Invertebrates

Product: EC 50 (Daphnia magna, 48 h): 2,18 mg/l

Components:
 2-ethylhexyl methacrylate EC 50 (Daphnia magna, 48 h): 2,18 mg/l
 Methyl methacrylate EC 50 (Daphnia magna, 48 h): 69 mg/l
 NOEC (Daphnia magna, 48 h): 48 mg/l

Toxicity to Aquatic Plants

Product: EC 50 (Selenastrum capricornutum (green algae), 72 h): 7,68 mg/l (OECD 201)

Components:
 2-ethylhexyl methacrylate EC 50 (Selenastrum capricornutum (green algae), 72 h): 7,68 mg/l (OECD 201)
 Methyl methacrylate EC 50 (Selenastrum capricornutum (green algae), 72 h): > 110 mg/l (OECD 201)

Toxicity to microorganisms

Product: EC 10 (activated sludge, 30 min): > 27,4 mg/l (OECD 209)

Components:
 2-ethylhexyl methacrylate EC 10 (activated sludge, 30 min): > 27,4 mg/l (OECD 209)
 Methyl methacrylate No data available.

Chronic hazards to the aquatic environment:

Fish

Product: No data available.

Components:
 2-ethylhexyl methacrylate No data available.
 Methyl methacrylate NOEC (Danio rerio, 35 d): 9,4 mg/l (OECD 210)
 LC 50 (Danio rerio, 35 d): 33,7 mg/l (OECD 210)

Aquatic Invertebrates

Product: NOEC (Daphnia magna, 21 d): 0,11 mg/l (OECD 211) Own study

Components:
 2-ethylhexyl methacrylate NOEC (Daphnia magna, 21 d): 0,11 mg/l (OECD 211)
 Methyl methacrylate NOEC (Daphnia magna, 21 d): 37 mg/l (OECD 211)
 EC 50 (Daphnia magna, 21 d): 49 mg/l (OECD 211)

Toxicity to Aquatic Plants

Product: NOEC (Selenastrum capricornutum (green algae), 72 h): 0,28 mg/l (OECD 201)

Components:
 2-ethylhexyl methacrylate NOEC (Selenastrum capricornutum (green algae), 72 h): 0,28 mg/l (OECD 201)
 Methyl methacrylate NOEC (Selenastrum capricornutum (green algae), 72 h): 110 mg/l (OECD 201)

Toxicity to microorganisms

Product: EC 10 (activated sludge, 30 min): > 27,4 mg/l (OECD 209)

Components:
 2-ethylhexyl methacrylate EC 10 (activated sludge, 30 min): > 27,4 mg/l (OECD 209)
 Methyl methacrylate No data available.

Persistence and Degradability

Biodegradation

Product:	88 % (28 d, OECD 301 C), Readily biodegradable
Components:	
2-ethylhexyl methacrylate	88 % (28 d, OECD 301 C) The product is easily biodegradable.
Methyl methacrylate	94 % (14 d, OECD 301 C) The product is easily biodegradable., aerobic

BOD/COD Ratio

Product:	No data available.
Components:	
2-ethylhexyl methacrylate	No data available.
Methyl methacrylate	No data available.

Bioaccumulative potential

Bioconcentration Factor (BCF)

Product:	Danio rerio, Bioconcentration Factor (BCF): 37 (OECD 305) Substance is metabolised via standard pathways. Therefore, significant bioaccumulation is not expected.
Components:	
2-ethylhexyl methacrylate	Danio rerio, Bioconcentration Factor (BCF): 37 (OECD 305) Substance is metabolised via standard pathways. Therefore, significant bioaccumulation is not expected.
Methyl methacrylate	No data available.

Partition Coefficient n-octanol / water (log Kow)

Product:	Log Kow: 4,95 20 °C (OECD 107)
Components:	
2-ethylhexyl methacrylate	Log Kow: 4,95 20 °C (OECD 107)
Methyl methacrylate	Log Kow: 1,38

Mobility in soil:

Product	The substance evaporates gradually into the atmosphere from the surface of the water. If the substance does get into the environment, it tends to remain in the compartment it was discharged into. Substance may bind to solid soil phase, sediment or clarification sludge due to its adsorptive behaviour.
Components:	
2-ethylhexyl methacrylate	The substance evaporates gradually into the atmosphere from the surface of the water. If the substance does get into the environment, it tends to remain in the compartment it was discharged into. Substance may bind to solid soil phase, sediment or clarification sludge due to its adsorptive behaviour.
Methyl methacrylate	No data available.
Product	This substance is not considered to be persistent, bioaccumulating and toxic (PBT). This substance is not considered to be very persistent and very bioaccumulating (vPvB).
Components:	
2-ethylhexyl methacrylate	No data available.
Methyl methacrylate	No data available.

Other adverse effects:**Other hazards****Product:**

Prevent substance from entering soil, natural bodies of water and sewer systems.

13. Disposal considerations**General information:**

Dispose of waste and residues in accordance with local authority requirements.

Disposal methods:

Waste is hazardous. It must be disposed of in accordance with the regulations after consultation of the competent local authorities and the disposal company in a suitable and licensed facility. The waste key number must be determined as per the European Waste Types List (decision on EU Waste Types List 2000/532/EC) in cooperation with the disposal firm / producing firm / official authority.

Contaminated Packaging:

Contaminated packaging should ideally be emptied; it can then be recycled after having been decontaminated. Packaging that cannot be cleaned should be disposed of professionally. Uncontaminated packaging may be taken for recycling.

14. Transport information**International Regulations****UNRTDG**

Not regulated as a dangerous good

IATA-DGR

Not regulated as a dangerous good

IMDG-Code

Not regulated as a dangerous good

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

15. Regulatory information**International regulations****Montreal protocol**

Not applicable

Stockholm convention

Not applicable

Rotterdam convention

Not applicable

Kyoto protocol

Not applicable

Inventory Status:

Australia AICS:	On or in compliance with the inventory	
Canada DSL Inventory List:	On or in compliance with the inventory	
China Inv. Existing Chemical Substances:	On or in compliance with the inventory	
Japan (ENCS) List:	On or in compliance with the inventory	
Korea Existing Chemicals Inv. (KECI):	On or in compliance with the inventory	
New Zealand Inventory of Chemicals:	On or in compliance with the inventory	
Philippines PICCS:	On or in compliance with the inventory	
Taiwan Chemical Substance Inventory:	On or in compliance with the inventory	Pre-registration is requested for specific importer.
US TSCA Inventory:	On or in compliance with the inventory	Commercial Status: Active
EINECS, ELINCS or NLP:	On or in compliance with the inventory	EU-REACH compliant for Evonik Operations GmbH and its affiliates as EU manufacturer/EU importer.

16. Other information, including date of preparation or last revision
Issue Date: 10.03.2019

Version #: 1.1

Further Information: The product is normally supplied in a stabilized form. If the permissible storage period and/or storage temperature is exceeded, the product may polymerize with heat evolution.

Revision Information: Changes since the last version are highlighted in the margin. This version replaces all previous versions.

Disclaimer: This information and all further technical advice is based on our present knowledge and experience. However, it implies no liability or other legal responsibility on our part, including with regard to existing third party intellectual property rights, especially patent rights. In particular, no warranty, whether express or implied, or guarantee of product properties in the legal sense is intended or implied. We reserve the right to make any changes according to technological progress or further developments. The customer is not released from the obligation to conduct careful inspection and testing of incoming goods. Performance of the product described herein should be verified by testing, which should be carried out only by qualified experts in the sole responsibility of a customer. Reference to trade names used by other companies is neither a recommendation, nor does it imply that similar products could not be used.

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